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Dissertation Title: *Essays in Macroeconomics*

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University Dissertation Fellowship, Yale University, 2026-2027
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Fall 2025, Teaching Assistant to Prof. Eduardo Davila, Undergraduate Financial Economics, Yale University

Spring 2025, Teaching Assistant to Prof. Ilse Lindenlaub, Undergraduate Intermediate Macroeconomics, Yale University

Fall 2024, Teaching Assistant to Prof. Zhen Huo, Undergraduate Intermediate Macroeconomics, Yale University

Summer 2023 and Summer 2024, Teaching Assistant to Prof. William Hawkins, PhD Math Camp, Yale University

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Research Assistant to Professor Ilse Lindenlaub, Yale University, 2023

Working Papers:

"Wage-Price Spirals: The Contract Accelerator," *Job Market Paper*

"From Technology Disruption to Adaptation: Optimal Policies for Taxation and Retraining"

"Occupational Skill Requirements, Separation Risk, and Skill Mismatch," with Matthew Famiglietti

"A Pendulum Model of Economic Growth," with Zhen Huo and Xiaodong Zhu

Seminar and Conference Presentations:

2026: Federal Reserve Bank of San Francisco (Brown bag seminar), Federal Reserve Bank of Richmond (Brown bag seminar), Midwest Macroeconomics Meetings, Chinese University of Hong Kong (Brown bag seminar), Hong Kong University of Science and Technology (Brown bag seminar)

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Dissertation Abstract

Endogenous Contract Choice and Wage-Inflation Spirals [*Job Market Paper*]

During inflation spikes, wage setting can become more responsive to inflation, either explicitly through contractual formulas that index wages to prices or implicitly through more frequent renegotiation. Linking wages to prices protects workers' purchasing power but transmits shocks more strongly into wages, costs, and prices, causing a feedback loop. While the amplification role of indexation has been widely studied, I study the interaction between wage-setting institutions and inflation dynamics: why does wage setting change with the inflation environment, and how do these changes reshape inflation dynamics? A wage-inflation spiral arises when wage indexation itself is endogenous and the pass-through coefficient of realized inflation to nominal wages responds optimally to the inflation environment.

Using contract-level records from a restricted Bureau of Labor Statistics archive of major U.S. collective bargaining agreements from 1974 to 1995, I measure whether and how strongly wages adjust automatically to prices within these multi-year contracts. Explicit cost-of-living adjustment (COLA) coverage among unionized workers rose to 60 percent in the 1970s and early 1980s, before falling back to 20 percent in the 1990s. COLA adjustment is nevertheless incomplete: a 1 percent increase in prices raises wages by about 0.85 percent on average.

To explain these facts, I build a model of contract choice and embed it in a New Keynesian framework. At signing, workers and a firm choose whether wages are indexed to inflation and, if so, by how much. COLA insures risk-averse workers against inflation surprises and can raise joint surplus relative to preset wages. Greater inflation volatility increases the gain from COLA and encourages adoption. However, full pass-through can be costly for financially constrained firms when revenues are weak, making partial indexation privately optimal. The economy's stock of indexed contracts becomes a state variable, which also makes the Phillips curve state-dependent: a higher COLA share makes marginal costs and inflation more responsive to aggregate shocks. In turn, this higher sensitivity raises inflation risk, making COLA more attractive and inducing strategic complementarity in adoption. In some cases, this complementarity can also support multiple self-fulfilling COLA equilibria. A more hawkish policy rule lowers inflation risk and COLA adoption, weakening the feedback; as uncertainty rises, stronger policy is needed to anchor wage setting.

I then test the model's central mechanisms and establish three causal relationships using identified shocks and instrumental variables: COLA adoption rises with uncertainty; inherited coverage raises subsequent adoption; and monetary policy shocks move inflation by more when coverage is high. Quantitatively, endogenous contract choice raises annualized inflation by 1.11 percentage points in 1973–75 and 0.75 percentage points in 1979–82—about 14 percent across the two episodes compared to the fixed-COLA counterfactual. It also doubles the contribution of the more hawkish Volcker-era policy rule to the more tranquil inflation by discouraging COLA adoption during the subsequent Great Moderation era.

I conclude that wage-setting institutions are not fixed background conditions: private insurance choices reshape inflation dynamics, while monetary policy can stabilize inflation partly by changing the contracts through which shocks are transmitted.

From Technology Disruption to Adaptation: Optimal Policies for Taxation and Retraining

Technological progress raises aggregate productivity but generates uneven gains across occupations, making redistribution and occupational transitions important policy concerns. This paper studies the joint optimal design of income taxation and retraining subsidies in response to technological disruption. I develop a two-sector Mirrlees framework with risky retraining outcomes and unobserved productivity. Productivity–training complementarity shapes both the efficiency gains from and the incentive costs of retraining. I estimate the retraining technology using German administrative records that track workers' retraining, employment, earnings, and occupations over time. I document strong productivity-training complementarity for short-term training. Quantitatively, under an automation shock, the constrained optimum prescribes short-term training for all manual workers, regardless of manual productivity. In contrast, laissez-faire underinvests in training due to uninsured outcome risk, and Germany's policy regime can misallocate workers in training choices due to misaligned incentives. At optimum, training expenditures amount to 0.65 percent of GDP, and moving from the actual to the optimal regime yields welfare gains equivalent to 0.33 percent of GDP.

Occupational Skill Requirements, Separation Risk, and Skill Mismatch, with Matthew Famigietti

Occupations differ in their job-loss risk. We document that job-loss risk is systematically related to occupational skill requirements. Conditional on worker attributes, occupations with greater manual requirements face higher separation rates, while those with greater cognitive requirements have lower separation rates. This appears to affect worker occupational choice. Using the NLSY, we document that workers with greater manual ability tend to hold jobs with higher cognitive requirements than those with lower manual ability, whereas workers with greater cognitive ability sort away from manually intensive occupations. We interpret these patterns through a dynamic model of multidimensional worker-occupation sorting with on- and off-the-job search, in which occupational requirements jointly determine match output and separation risk. The correlation between skill requirements and job-loss risk rotates job-acceptance sets, inducing workers to sacrifice match quality for job security. In the calibrated economy, aggregate output is 0.66 percent

lower than in a counterfactual with homogeneous separation risk and the same aggregate employment-to-unemployment transition rate. The decentralized economy also produces 0.56 percent less output than the planner's allocation because workers respond more strongly to separation risk. A budget-neutral unemployment-insurance policy targeted by worker ability improves occupational sorting and recovers nearly half of this output gap.

A Pendulum Model of Economic Growth, with Zhen Huo and Xiaodong Zhu

This paper develops a "pendulum model" of economic growth to explain the recurrent oscillations between state control and market liberalization observed in major economies. We posit that policymakers face a fundamental trade-off between promoting growth and maintaining political or systemic control over the accumulation of private wealth or capital. Using a dynamic framework where a planner balances these competing objectives, we show that high growth targets endogenously generate cycles: periods of pro-market expansion are necessary to meet growth targets but inevitably increase the share of private resources, triggering a retrenchment phase to restore control. We provide empirical evidence supporting this mechanism, focusing primarily on China's post-reform history, where policy swings between "pro-private" and "pro-state" regimes align closely with the model's predictions. To demonstrate the broader applicability of the trade-off between efficiency and stability, we present supplementary evidence from the United States, illustrating similar cyclical patterns in financial regulation and credit cycles. Our findings suggest that the volatility of economic policy is not merely a response to exogenous shocks but an inherent feature of governance under binding growth constraints.